

F1 Analytics & Prediction Platform — Task Tracker

Phase 0: Project Setup & Data Infrastructure

- ☒ Create `requirements.txt`
- ☒ Install dependencies (pip install)
- ☒ Create `src/__init__.py`
- ☒ Create `src/data_loader.py`
- ☒ Create `src/feature_engineering.py`
- ☒ Create `src/models.py`
- ☒ Create `src/visualization.py`
- ☒ Create `.gitignore`

Phase 1: Data Exploration (Notebook 01)

- ☒ Create `notebooks/01_data_exploration.py`

Phase 2: Race Winner Prediction (Notebook 02)

- ☒ Create `notebooks/02_race_winner_prediction.py`

Phase 3: Driver Performance Analysis (Notebook 03)

- ☒ Create `notebooks/03_driver_performance_analysis.py`

Phase 4: Strategy Optimization (Notebook 04)

- ☒ Create `notebooks/04_strategy_optimization.py`

Phase 5: Championship Prediction (Notebook 05)

- ☒ Create `notebooks/05_championship_prediction.py`

Phase 6: GOAT Ranking (Notebook 06)

- ☒ Create `notebooks/06_goat_ranking.py`

Phase 7: Driver Clustering (Notebook 07)

- ☒ Create `notebooks/07_driver_clustering.py`

Phase 8: Race Simulation (Notebook 08)

- ☒ Create `notebooks/08_race_simulation.py`

Phase 9: Constructor Prediction (Notebook 09)

- ☒ Create `notebooks/09_constructor_prediction.py`

Phase 10: Interactive Dashboard

- ☒ Create `dashboard/app.py`
- ☒ Create dashboard pages (6 pages)
- ☒ Fix `set_page_config` conflicts across pages

Verification

- ☒ Run all scripts end-to-end
- ☒ Launch and verify Streamlit dashboard (running on port 8501)
- ☒ Create walkthrough with explanations